discretize $z = \log(y)$, 25 for $\widehat{f}$, and 45 for B, for a total of 78,750 points. To solve the four HJB equations, we set the following boundary conditions:

- on $z_{\min}$: $v_z^{def,\infty} = v_z^{\infty} = w_z$, where $w = w(z)$ is the solution of a related problem where the country is in default without the possibility to get back to issue new debt, without pollution and disasters. The motivation of this assumption is that, if the output is close to zero, the economic conditions are so bad that the country immediately defaults when the exclusion period ends, and pollution and climate disasters have no effect on such a bad economic condition. We also set $Q_z^{def,\infty} = Q_z^{\infty} = 0$, as in [32];
- on $z_{\max}$: $v_z^{def,\infty} = v_z^{\infty} = 0$, as in [32], and $Q^{def,\infty} = \frac{\theta_D}{1+\delta/\chi}$, $Q^{\infty} = 1$, i.e., the output is so high that the country, in case of default will not default anymore at the end of the exclusion period, and, in case of default it opts not to default, i.e., the country is default-free;
- on $\widehat{f}_{\min}$ and $\widehat{f}_{\max}$: zero second order derivative with respect to $\widehat{f}$ for all the four functions;
- on $B_{\max}$: $v^{\infty} = v^{def,\infty}(\eta y)$, $Q^{\infty} = Q^{def,\infty}(\eta y)$, the debt is so high that we assume that the country defaults.

No boundary conditions are needed on $B_{\min}$, $B_{\max}$ for the def functions, since there is no derivative with respect to B, and on $B_{\min}$ for the other two functions due to the use of the upwind scheme and the absence of the second order derivative with respect to B.

Once the asymptotic solutions are computed, we set these solutions as terminal condition at a time T, with T chosen such that $k_e e^{-gT} < 10^{-5}$, that is, the role of the time in the coefficient of the HJB equations is negligible, and we solve backward in time with 500 time steps, to compute the solutions at time 0 and all the optimal controls. We exploit the same boundary conditions above described.

C Sensitivity analysis

In this section we report the sensitivity plots for the SA-APAC region. Plots for the others regions are available upon request.

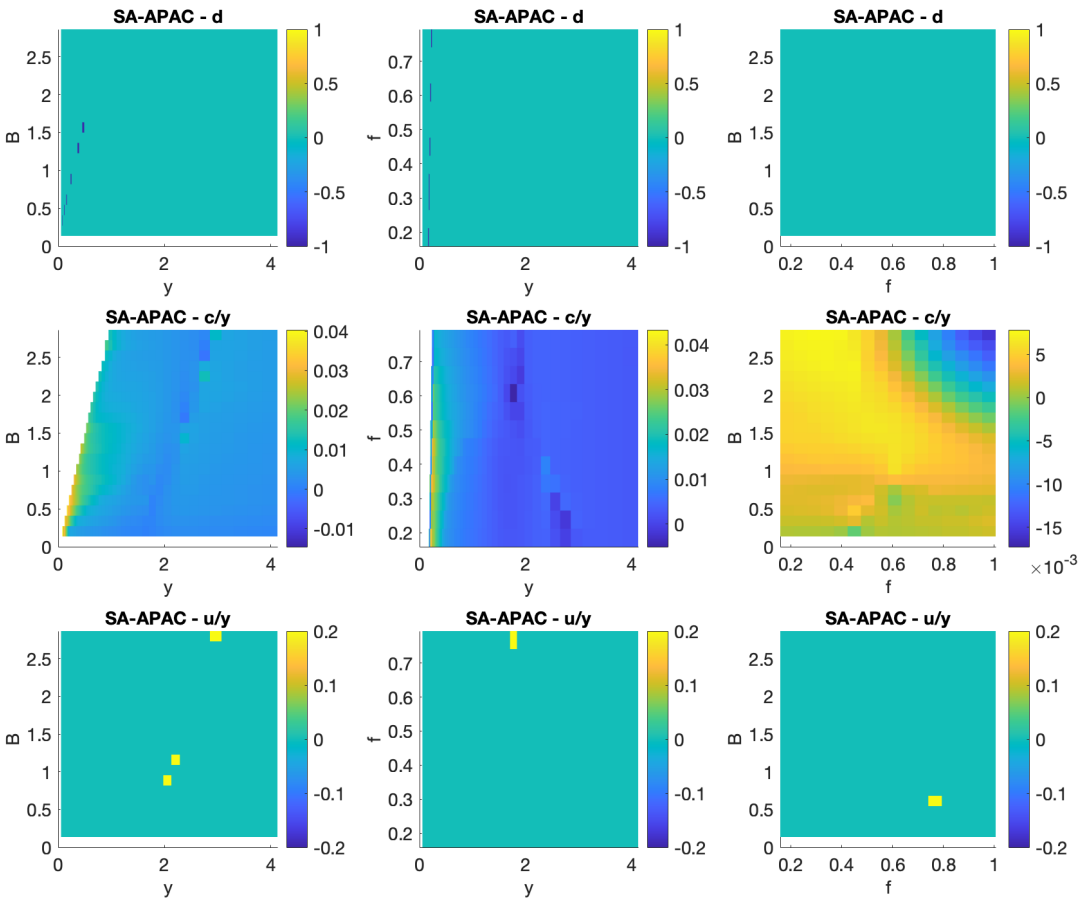

Figure C.1: Sensitivity analysis - parameter χ, SA-APAC region. We report the differences of the optimal controls value obtained with $\chi = 1.2\chi_b$ and $\chi = 0.8\chi_b$, χ_b being the baseline value of the parameter, that is, if the decision to default (d) is considered, we plot $d_{1.2\chi_b} - d_{0.8\chi_b}$. See Figure 4 for details and for the plots of the baseline value.

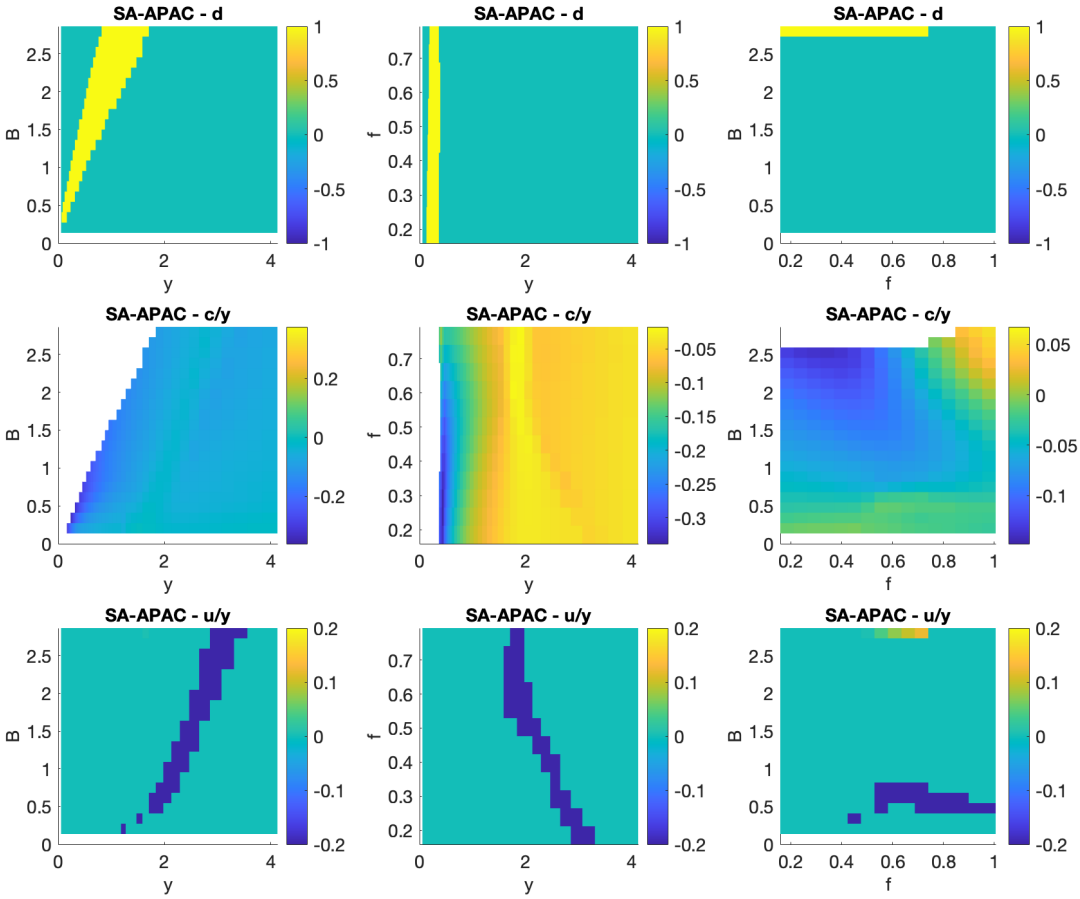

Figure C.2: Sensitivity analysis - parameter η, SA-APAC region. We report the differences in $d,\ c/y,\ u/y$ obtained for $\eta = 1.2\eta_b$ and $\eta = 0.8\eta_b$, η_b being the baseline value of the parameter, in case of d we plot $d_{1.2\eta_b} - d_{0.8\eta_b}$, see Figure 4 for details and plots of the baseline parameter.